

Does this product have real-world validity?

Roofing, solar, and adjacent industries | Burlington, Vermont focus

Bottom line: drone-based roof measurement is a proven, defensible product in one place, solar, where precision has money attached. In commodity roofing you cannot win: incumbents give away instant measurement reports for \$13-38 with no site visit, and a drone job costs \$150-400 just to show up. Your validated-accuracy edge is real, but it only pays where accuracy is worth a premium: solar design, complex or unmapped roofs, and defensible inspection. In Burlington, as a part-time solo operator, this is a good side income, not a company.

Solar best-fit vertical	\$5-35k/yr realistic SOM (side project)	Side income not a venture, as configured
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What you are holding

A decision memo, not research theater. It ranks seven candidate verticals by fit, shows why commodity roofing is a price trap, sizes the Burlington opportunity bottom-up, resolves the Tyco question with evidence, and states the downside cases. Every figure is sourced on the final page or labeled an estimate.

How to read the recommendation

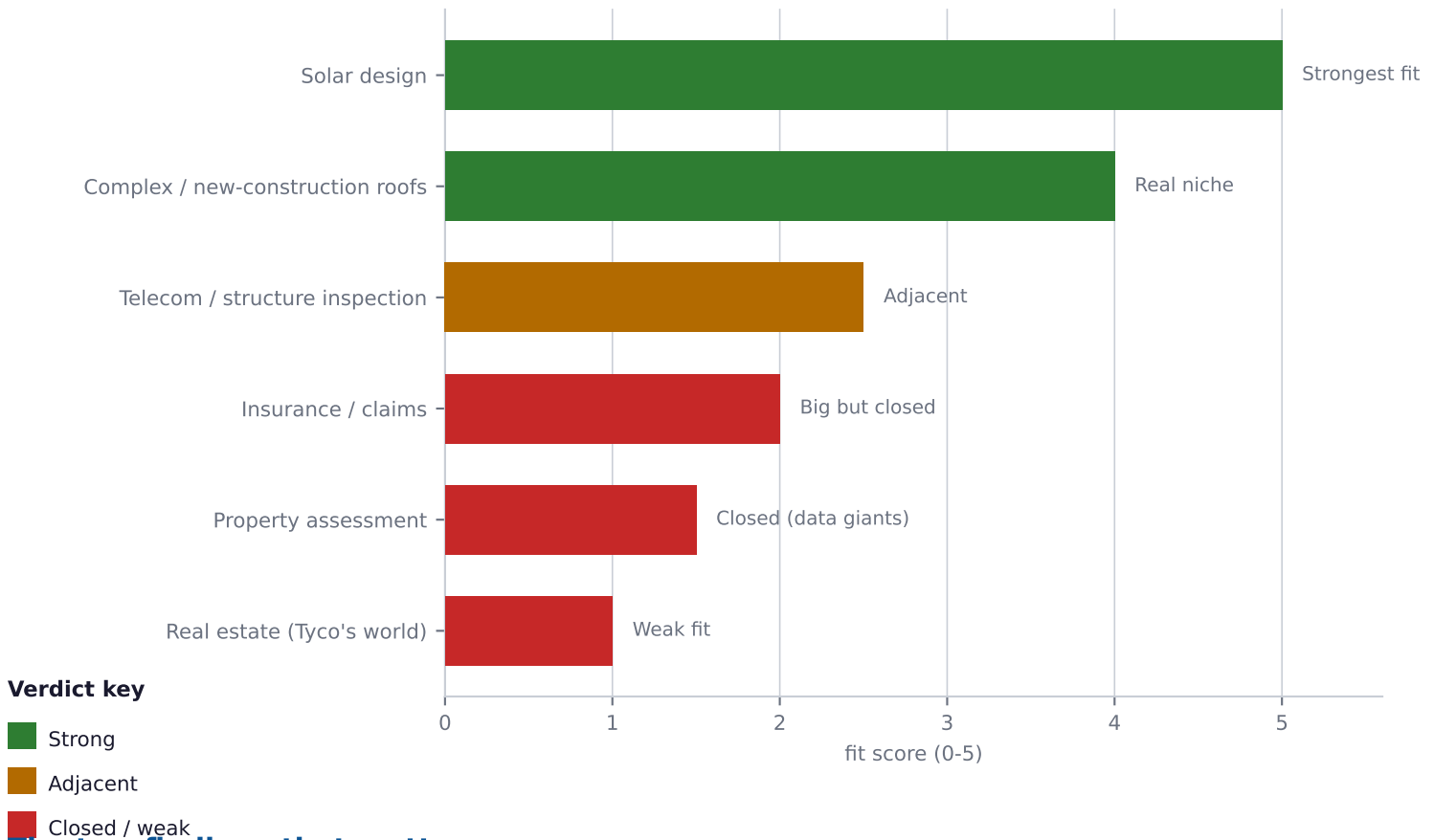
Three goals were in play: is it a real business, which vertical wins, and an honest validity check. The answer differs by goal, so the recommendation on p.5 is split three ways rather than forced into one verdict.

Scope note

Measurement-as-a-service is the base case (you fly, sell a report per job). The national contractor-kit model is treated as an upside scenario, gated by the Part 107 adoption barrier. Software-for-installers is noted, not anchored on.

Vertical ranking (best fit first)

Fit is a judgment score (0-5) combining how much the buyer values measurement precision against how open the market is to a solo local operator. Bars are colored by verdict, not by rank.



The two findings that matter

1. Solar is the one vertical where drone photogrammetry already won. Scanify, effectively the company version of this pipeline, sells 1-3 inch drone-derived solar designs because panel fit, shading, and lender bankability put real money on precision. Vermont's strong net-metering incentives keep local solar demand alive.

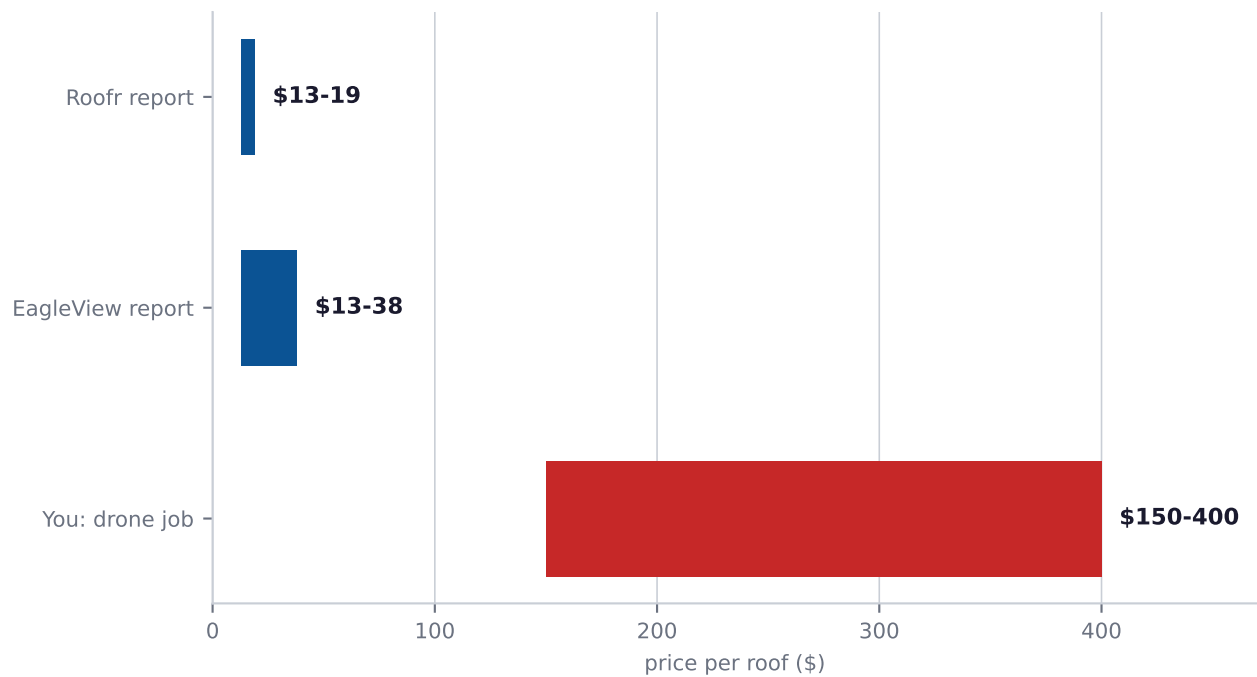
2. The vertical this project was born from, real estate, is the worst fit. Real estate buyers want imagery and video, not roof measurements. Your product does not serve Tyco's customers' actual need. That sounds like bad news but it dissolves the conflict: you are not competing with his business (see p.5).

Why the closed markets are closed

Insurance, claims, and property assessment are enormous and measurement-hungry, but already owned by aerial-data giants: Cape Analytics, Vexcel, and Nearmap sell property intelligence to all of the top-10 P&C carriers at national scale. That is an enterprise data business, not a job a certified solo pilot in Burlington can win. Telecom and structure inspection grow fast (~16%/yr) but are an inspection market needing different skills, not a measurement wedge for this pipeline.

Why commodity roofing is a price trap

This one chart is the whole game. For a simple roof, incumbents deliver a measurement report from existing imagery in minutes for the price of lunch. Your drone has to physically show up. You are 5-10x the price and slower, and that gap is the market telling you the commodity report is a solved problem.



Competitive landscape

Player	Method	Price	Speed	Site visit?
Roofr	Satellite imagery	\$13-19 / report	< 3 hr	No
EagleView	Aerial imagery library	\$13-38 (to \$87)	Min-hours	No
Hover	Phone photogrammetry	Free / \$149+/mo	Same day	Homeowner
Scanify	Drone photogram. (solar)	\$1,499 / yr sw	Same day	Drone
You (MaaS)	Drone photogrammetry	\$150-400 / job	Same/next day	Drone

The takeaway is not that your product is worse. It is that the drone visit only pays when it delivers something the \$13 report cannot: solar-grade precision, a roof the satellite cannot see (new construction, heavy tree cover, very steep or tall), or an inspection with liability attached. Sell those jobs. Never try to beat \$13 on a simple suburban roof, because you cannot.

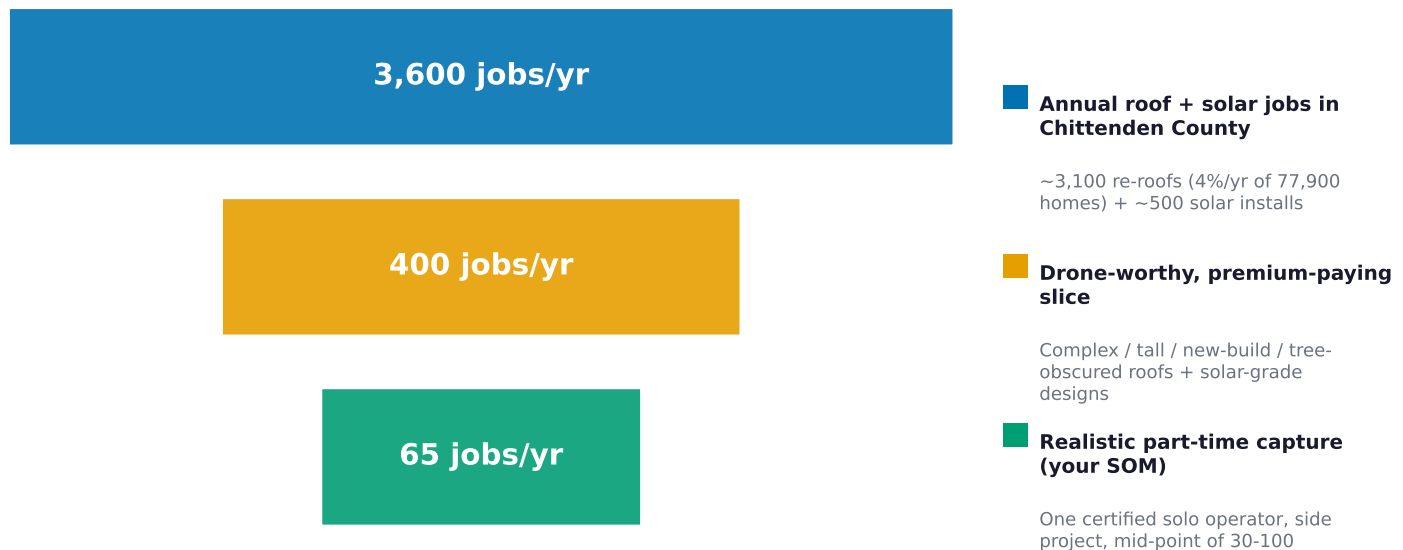
Note the incumbents are also moving: EagleView One and Hover's 2026 relaunch both add 3D building models. If they close the complex-roof gap, the wedge narrows. This is tracked as a risk on p.6.

How big is it, honestly

TAM is national context; the number that matters to you is the bottom-up local funnel. The three tiers are in different units of scale (a national niche vs one part-time operator), so they are shown as tiles, not a single misleading funnel.



Bottom-up: annual jobs in Chittenden County



Assumptions you confirmed

Two levers drive the SOM: a ~4%/yr roof-replacement rate (a 20-25 year roof life) applied to Chittenden's 77,900 housing units, and ~30-100 jobs/yr of realistic part-time capacity for one certified operator. At \$150-350 per drone job, that lands at roughly \$5k-35k/yr gross, low five figures net after drone, insurance, software, and Part 107 costs.

The national TAM (~\$0.2-0.9B/yr for drone-based roof/property measurement and inspection) confirms the industry is real and growing 14-19%/yr. It does not change the local reality: as a side project, the ceiling is a good second income, and the throughput cap is you, one pilot with a Part 107.

The Tyco question, answered with evidence

The research resolves this more easily than you feared. There is no conflict, because Tyco's customers do not want what you built. Real estate buyers want video and stills, not roof measurements. So you are not secretly competing with his business or repurposing his clients.

The real synergy with Tyco is not the product, it is the shared infrastructure: he already holds a Part 107, flies a drone, has capture skill and local reputation. That makes him a natural capture partner or referral source (he flies or refers the solar and complex-roof leads he cannot serve), not a customer and not someone you are deceiving. The honest framing costs you nothing to say out loud: you are building an adjacent thing that can ride alongside his operation, not a thing for him.

Recommendation, split by your three goals

If the goal is income now

Run it as a solar-and-complex-roof drone service in Chittenden County, not a general roof-measurement service. Skip commodity roofing entirely. Partner loosely with Tyco for capture and referrals, and tell him plainly what it is.

If the goal is a real business

The only version that scales is Scanfly's model, software for solar installers, and you would be entering behind a funded incumbent. Worth knowing; probably not worth doing as a student side project.

For the resume / interview goal

This analysis is itself the asset. 'I built a validated-accuracy pipeline, then found through market analysis that the defensible market was solar, not the roofing/real-estate use I started from' is a strong, honest engineering-and-judgment story.

The one-line strategy

Sell precision where precision is worth paying for (solar, complex roofs, defensible inspection), around Burlington, part-time, with Tyco as a capture ally. Do not compete on price, speed, or simple roofs; those belong to the imagery incumbents and always will.

Risks and counterarguments

Incumbents move down-market

EagleView One and Hover's 2026 relaunch add 3D building models. If they close the complex-roof gap, your niche shrinks.

Solar is in a cyclical dip

US residential solar fell ~31% in 2024 on high interest rates and net-metering cuts. Your best vertical is soft right now.

Part 107 caps throughput

The certification is a moat against competitors, but also the reason the SOM is small: one part-time pilot has a hard capacity ceiling.

The scalable path is the one you dislike

Scanify proves the winning model is software sold to installers, not measurement-as-a-service. The venture-scale version is the software play you set aside.

Method & honesty note

Sizing uses top-down market reports for context and a bottom-up job count for the local reality; every leap is labeled. Figures are as of July 2026 and will drift. This is a decision memo, so it commits to a recommendation rather than hedging, and states its downside cases above rather than burying them.

Sources

Claim	Source
EagleView pricing	getapp.com/construction-software/a/eagleview/pricing
Roofr pricing update 2025	roofr.com/product-blog (pricing update)
EagleView vs Hover	roofingsoftwareguide.com/comparisons/eagleview-vs-hover
Drone vs satellite accuracy	airteam.ai (2025 drone/satellite/aircraft)
Scanify accuracy	scanify.com/blog/how-accurate-is-scanify-software
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Cape Analytics + Vexcel	prnewswire.com (Cape-Vexcel imagery partnership)
US roofing market size	mordorintelligence.com (United States roofing)
Chittenden Co. housing (77,866 units)	censusreporter.org (Chittenden Co, VT)
VT solar net metering	solarpowerworldonline.com (VT net metering 2024)
US residential solar -31% in 2024	pv-magazine.com (2025-03-14)
Drone roof inspection \$150-400/job	homeguide.com (drone roof inspection cost)
Drone tower inspection market	factmr.com (drone tower inspection)